

Here are just the problems from the final last spring.

Spring 2021 Final Exam - 4pm Test

⚠ This is a preview of the published version of the quiz

Started: Dec 7 at 8:22am

Quiz Instructions

First of all, if you have technical difficulties with your computer, software, or internet access, **DO NOT WORRY**. Restart your browser and try again. That usually solves the problem. If you worry you will be pressed for time and want to add a few minutes that you lost because of the restart, email David Burk (david.burk@georgetown.edu (<mailto:david.burk@georgetown.edu>)). If restarting doesn't solve the issue, email David Burk. Please do NOT worry. We will make it work. We will treat you fairly. We will make it right!

Once you open the exam, you have **120 minutes** to take it.

This exam has 130 points. 20 problems are standalone multiple choice problems. There are three "multi-question problems" that are a mix of numerical answer and multiple choice.

The exam is open-book and open-notes. You may even use Google, etc. But you MAY NOT communicate with any person during the exam.

You can and should have scratch paper on which you can write to help you solve some problems.

You may not use an electronic device other than your laptop. No use of phones, tablets, etc., is allowed.

If you can't get a problem in a couple minutes, go on to the next one.

Do not worry if you think the exam is hard; remember, other people may think it is hard too, in which case it would be curved. Just keep going and do your best.

If you think there is an error in the exam, do not worry. During the exam, DO NOT reach out to me or a TA about errors in the exam: We will not respond. (Because we can't communicate with all of you about problems on the exam we can't fairly communicate with ANY of you). Just choose the best possible answer. If there is indeed an error, we will adjust your score after the exam.

Good luck!

Question 1**5 pts**

For a firm with market power, the marginal revenue curve

- ☐ lies below the demand curve.
- ☐ lies below the average revenue curve.
- ☐ could be below zero for some quantities (that is, at some levels of production MR is negative).
- ☐ All of the above

Question 2**5 pts**

You sell burritos on campus. When you charge \$6 / burrito you sell 20 burritos. When you charge \$5 / burrito, you sell 25 burritos. Your total revenue increases because

- ☐ the output effect outweighs the discount effect.
- ☐ the output effect and the discount effect both push total revenue up.
- ☐ the discount effect outweighs the output effect.

- ☐ the demand curve you face slopes down.

Question 3

5 pts

A drug company has a monopoly on a vaccine for a certain virus that is not lethal but does make people sick enough to miss work or school for a few days. Because this vaccine is supplied by a monopolist,

- ☐ less than the socially efficient quantity (that is, less than the quantity that maximizes economic surplus in society) of the vaccine is sold.
- ☐ more than the socially efficient quantity of the vaccine is sold.
- ☐ the firm sets its price equal to its marginal cost.
- ☐ there is no reason to advertise.

Question 4

5 pts

If a firm with market power can perfectly price discriminate, then the firm's MR curve is

- ☐ the same as the demand curve the firm faces.
- ☐ a horizontal line.
- ☐ beneath the demand curve the firm faces.
- ☐ also its marginal profit curve.

Question 5

5 pts

A firm with market power will

- ☐ face a downward sloping demand curve.
- ☐ might be a price-taker if it sells an undifferentiated good.
- ☐ makes short-run decisions based on fixed costs (such as rent).
- ☐ All of the above

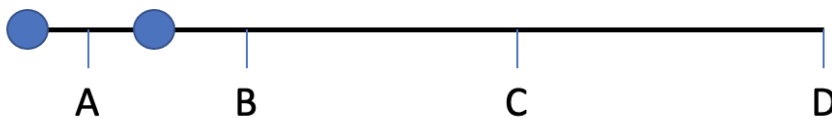
Question 6**5 pts**

There are only two grocery stores in your city. They are located across the street from one another and sell the exact same goods. If one grocery store closed (assume there was no change in the market demand for groceries), you would expect the remaining grocery store to

- ☐ increase its prices.
- ☐ would increase the quantity of groceries it sells, up to the quantity that maximizes total economic surplus in your town.
- ☐ keep its price the same so as to not irritate its loyal customers.
- ☐ lower its price in order to capture all the demand in your city.

Question 7**5 pts**

Suppose you are opening a hot cocoa stand on a street that already has two hot cocoa stands, as represented by the circles in the picture below. Potential customers only choose a hot cocoa stand based on which is closest. Suppose that customers are spread evenly along the street when they are suddenly struck by the desire for hot cocoa. Which one of the following four spots should you place your hot cocoa stand? (Assume that no one else will open a new cocoa stand after you.)



- ☐ A
- ☐ B
- ☐ C
- ☐ D

Question 8**5 pts**

Suppose on January 1 a firm rented a factory and some machines for the entire year (suppose the firm paid the entire year's rent on January 1). In addition to those fixed costs, each week the firm would hire as many workers as it needed for the week (i.e., the workers were a variable cost).

In July, the firm realizes that, if it stay in business through the end of the year, its total cost will be greater than its total revenue. That is, it would have negative economic profits if it stays in business the whole year.

Then

- ☐ the firm should stop producing—i.e., stop hiring workers and stop making goods—as soon as possible.
- ☐ the firm should keep producing this year since it already paid rent for the entire year.
- ☐ the firm should keep producing this year if it can produce at a point where its MR is greater than its MC; otherwise, it should stop producing immediately.
- ☐ the firm should lower its price to sell more goods and make more money.

Question 9

5 pts

If Bryan's wealth goes up \$10 from its current level, he will get a utility boost of 4 "points". If his wealth goes down by \$10, he would lose 6 points of utility.

Will Bryan, a very rational individual, take a bet that would win him **\$20** with probability 50 percent and lose him \$10 with probability 50 percent?

- ☐ Yes, because the expected payout of that bet is positive.
- ☐ Yes, because his expected utility if he takes the bet is higher than his expected utility if he doesn't take the bet.
- ☐ No, because the expected utility of the bet is negative.
- ☐ Can't say, because of diminishing marginal utility of wealth.

Question 10

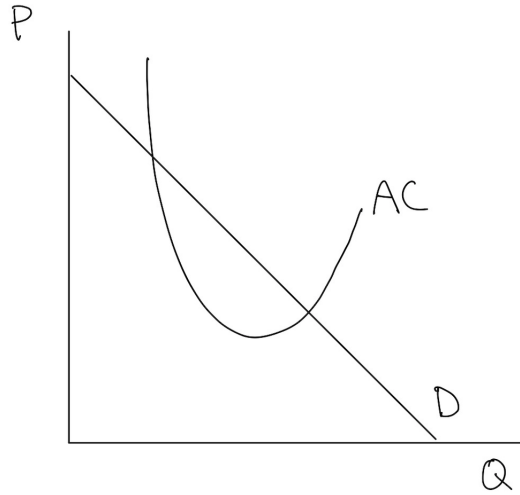
5 pts

Consider the imperfectly competitive market for cupcakes in the Georgetown neighborhood of the great world city of Washington, DC, USA. Suppose in 2020 there was a substantial decrease in the DC market demand for cupcakes, causing several cupcake firms to exit the market at the end of 2020. Suppose there were no further changes in the DC market demand for cupcakes (so the market demand for cupcakes stayed at its reduced, 2020 level). Then remaining firms in the market would each

- ☐ a) see an increase in profits relative to 2019 profits because their "firm demand curves" (i.e., the demand curves their firms face) shift out as the result of the exit of rivals.
- ☐ b) see a decrease in profits relative to 2019 because of the reduced market demand.
- ☐ c) have the same profits, because there's no reason their MC or MR curves would shift.
- ☐ d) can't say without more information.

Question 11**5 pts**

Suppose a profit-maximizing firm faces the demand curve shown and has the average cost curve shown.



Then

- ☐ a) the firm is earning economic profits.
- ☐ b) the firm should produce at the quantity that maximizes the gap between the demand curve and the AC curve (i.e., that maximizes $AR - AC$)
- ☐ c) the firm produces at a quantity where the AC curve crosses the demand curve.
- ☐ a and b
- ☐ None of the above.

Question 12**5 pts**

To start your job search, a career advisor recommends you create a LinkedIn profile. With LinkedIn, you can advertise your skills, share previous work experience, and connect with recruiters. You don't like the user interface of the LinkedIn website, but you won't change to a different job search site because so many prospective employers only use LinkedIn. Of the following, which is the most effective source of an entry barrier that LinkedIn has established.

- ☐ reputation and goodwill
- ☐ network effect
- ☐ economies of scale
- ☐ switching costs

Question 13**5 pts**

You manage a restaurant and buy vegetables from a local farm. When you see another local farm for sale, you realize it could provide the vegetables you need. If you buy the farm, which of the following is an advantage of that vertical integration?

- ☐ you'll get better vegetables
- ☐ you'll avoid the hold-up problem with your vegetable supplier
- ☐ it will raise the prices the original farm charges
- ☐ all of the above

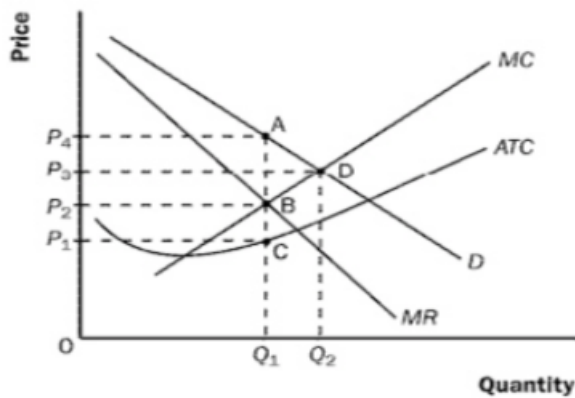
Question 14**5 pts**

Firms engage in advertising in order to

- ☐ a) reduce their costs.
- ☐ b) increase the demand for their products.
- ☐ c) make the demand for their product less elastic.
- ☐ d) make the demand for their product more elastic.
- ☐ b and c
- ☐ b and d

Question 15**5 pts**

In the figure below, which expression represents the **profit** of a profit-maximizing firm with market power? [Note: ATC is "average total cost" or what your textbook just calls "average cost"]



- ☐ $P_4 \times Q_1$
- ☐ $(P_4 - P_1) \times Q_1$
- ☐ $(P_4 - P_2) \times Q_1$
- ☐ $P_3 \times Q_2$
- ☐ Can't say with this information.

Question 16

5 pts

Your favorite fast food restaurant charges \$3 for a standalone drink, but only \$1 dollar for the drink if you add the drink to a meal. Of the following, what is **the most likely** explanation for why the restaurant prices the drink that way?

- ☐ The drink is cheaper for the restaurant to prepare and serve if they're already preparing a meal for you.
- ☐ The restaurant is trying to develop a good reputation with customers so gives them the drink at below marginal cost, so long as they buy a meal.
- ☐ the restaurant is trying to increase its profits through this form of price discrimination.
- ☐ The restaurant prefers to sell food to drinks so discourages people from buying just drinks by raising the price of drinks.

Question 17

5 pts

Suppose that there are two types of potential customers for your product: those who have a reservation price of \$10 and those who have a reservation price of \$5. If you can't price discriminate, what price should you charge?

- ☐ a) \$5, to ensure the maximum number of customers.
- ☐ b) Something between \$5 and \$10, to increase your total revenue.
- ☐ c) It depends on the number of demadners of each type.

☐ d) It depends on your MC curve.

☐ c and d

Question 18

5 pts

Suppose a movie theater was charging \$10 per ticket for all customers but last week it tried a new, two-price system where it charges \$12 per ticket for non-students and \$8 per ticket for students. ***(It picked the new prices based on a guess, not on a careful understanding of demand.)*** Which one of the following outcomes is most likely to be true?

☐ total consumer surplus would increase

☐ total producer surplus would increase

☐ the total consumer surplus of students would increase and the total consumer surplus for non-students would decrease

☐ total consumer surplus for students would increase and total consumer surplus for non-students would either increase or decrease.

Question 19

5 pts

A person who is risk averse will

☐ a) prefer a certain \$1,000 to a bet that has an expected financial value of \$1,000 (assuming that the person doesn't get any entertainment value from gambling)

☐ b) have a utility curve that slopes downward

☐ c) have diminishing marginal utility of wealth.

☐ a and c

Question 20

5 pts

You are a teacher with a bachelor's degree and you make \$40,000 a year. Your living expenses are \$15,000 a year.

If you had a master's degree you would get paid \$50,000 a year for the same job.

The closest master's program has a tuition of \$15,000 each year for two years. To do the masters program you would have to temporarily quit your job and move to the neighboring town, where your living expenses would be \$20,000 a year for each of the two years of the program. (After the program you would move back to the original

town, where living expenses were only \$15,000. And you would go back to your old job, now with the higher salary.)

Assume that your quality of life is the same whether working or in graduate school and the same in both towns. Assume all the relevant costs and benefits have been stated above.

If you chose to do the master's program, What is the minimum number of years you would have to work at that job after doing the master's program to justify the **total economic cost** of getting the master's degree? (Hint: Think about this problem like an accounting profit vs. economic profit problem.)

☐ 4 years

☐ 8 years

☐ 12 years

☐ 13 years

☐ 14 years

Question 21

3 pts

This is the first part of a multi-question problem.

You manage a firm with market power. The demand curve that your firm faces is

$Q = 10 - P$. The marginal revenue curve your firm faces is $MR = 10 - 2Q$.

Suppose your fixed costs were 10 and your marginal cost curve is given by $MC = 2$.

a) What is the **price** you set if your goal is to maximize the firm's profits? [Write just the number. Do not write "\$"]

Question 22

3 pts

This is the second part of a multi-question problem. Refer to the information in the previous part.

b) What is your firm's total profit? [Write just the number; no "\$".]

Question 23**3 pts**

This is the third part of a multi-question problem. Refer to the information in the first part.

c) Suppose now that your firm could perfectly price discriminate. How many units would it sell?

Question 24**3 pts**

This is the first part of a multi-question problem.

Rafa's wealth is \$10,000 when a friend comes to him with two investment opportunities.

Investment A would cost Rafa \$1,000. With 75 percent probability his \$1,000 investment would grow to \$2,000. With 25 percent probability, it would shrink to 0.

Investment B would cost Rafa \$2,000. With 50 percent probability the investment would grow to \$6,000 and with 50 percent probability it would shrink to 0.

a) What is the expected financial gain from Investment B? (Note, I am not asking for Rafa's expected wealth. I am asking for the expected financial gain from the investment).

Question 25**3 pts**

This is the second part of a multi-question problem. Refer to the information in the previous part.

b) Suppose Rafa can invest in at most one option. And he may of course choose not to invest at all. What will Rafa choose?

☐ Investment A

☐ Investment B

☐ Don't invest.

☐ Insufficient information to know what Rafa will do.

Question 26**3 pts**

This is the third part of a multi-question problem. Refer to the information from the previous parts (but do not use information from the next part to answer this question).

c) Here's some additional information: Rafa has diminishing marginal utility of wealth. With this additional information, what will Rafa choose?

- ☐ Investment A
- ☐ Investment B
- ☐ Don't invest.
- ☐ Insufficient information.

Question 27**3 pts**

This is the fourth part of a multi-question problem. Refer to the information from the previous parts.

d) Here's some additional information: Rafa's utility over various income levels is given in the following table.

Wealth (\$)	Utility
8,000	4
9,000	8
10,000	11
11,000	12
12,000	12.5
13,000	12.75
14,000	13
15,000	13.125
16,000	13.2

What is Rafa's expected utility from **Investment B**?

Question 28**3 pts**

Your town is going to connect the highway to the town center by building two ramps. Exxon and Shell are both interested in building a single gas station. They can build the gas station near the ramp for northbound traffic or they could build the gas station near the ramp for southbound traffic. The profits, in thousands of dollars, for each outcome are illustrated in the following payoff table:

	Shell builds near northbound ramp	Shell builds near southbound ramp
Exxon builds near northbound ramp	Exxon receives \$60 thousand profit Shell receives \$50 thousand profit	Exxon receives \$110 thousand profit Shell receives \$100 thousand profit
Exxon builds near southbound ramp	Exxon receives \$80 thousand profit Shell receives \$120 thousand profit	Exxon receives \$50 thousand profit Shell receives \$50 thousand profit

What is the outcome that has the highest producer surplus?

- ☐ a) "top-left" (i.e., both build near north ramp)
- ☐ b) "top-right" (Exxon builds north; Shell builds south)
- ☐ c) "bottom-left" (Exxon builds south; Shell builds north)
- ☐ d) "bottom-right" (i.e., both build near south ramp)
- ☐ c and d

Question 29

3 pts

Don't worry about this problem! We didn't cover that much game theory.

Refer to the previous game. Suppose the two firms don't communicate and must choose at the same time (i.e., it's a simultaneous game). Identify the Nash equilibrium (or equilibria).

- ☐ a) "top-left"
- ☐ b) "top-right"
- ☐ c) "bottom-left"
- ☐ b and c
- ☐ there is none

Question 30**3 pts**

Don't worry about this problem; we didn't cover that much game theory.

Refer to the Exxon-Shell game of the previous two problems.

What would happen if Shell could move first (that is, if this were a sequential game and Shell moved first)?

- ☐ a) "top-left"
- ☐ b) "bottom-right"
- ☐ c) "bottom-left"
- ☐ d) "top-right"
- ☐ b OR d; can't say without more information

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